DATA SCIENCE PROJECT

Household Energy Consumption Analysis

PROJECT PREPARED BY

Data Cleaning

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Household Energy Consumption Analysis

Project Overview

4 Years
SPAN (DEC 2006 - NOV 2010)

1-Min
GRANULARITY LEVEL

2007–2010
POWER DATASET

This project cleans, engineers features, and analyzes the **Individual Household Electric Power Consumption** dataset (spanning December 2006 to November 2010).

The objective is to prepare minute-by-minute electrical measurements for time-series modeling and uncover meaningful patterns in household electricity usage across different time scales and appliance groups.

Household Energy Consumption Analysis

Dataset Description & Schema

COLUMN NAME	UNIT / FORMAT	DESCRIPTION
Datetime	YYYY-MM-DD HH:MM:SS	Timestamp of the recording (sampled every minute).
Global_active_power	Kilowatts (kW)	Total active power consumed by the household.
Global_reactive_power	Kilowatts (kW)	Total reactive power consumed by inductive/capacitive loads.
Voltage	Volts (V)	Average minute-by-minute household AC voltage.
Global_intensity	Amperes (A)	Average current intensity drawn by the household.
Sub_metering_1	Watt-hours (Wh)	Energy consumed by the **kitchen** (dishwasher, oven, microwave).
Sub_metering_2	Watt-hours (Wh)	Energy consumed by the **laundry room** (washing machine, dryer, fridge, light).
Sub_metering_3	Watt-hours (Wh)	Energy consumed by **climate control** (water heater, AC).

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Project Workflow & Notebooks

1 01_Data Cleaning

- Loads raw dataset (**2,075,259 rows**, 9 cols).
- Merges **Date** & **Time** to Datetime; converts power metrics to float64.
- Removes **25,979 missing records**; verifies zero duplicates.
- Exports clean dataset (**2,049,280 rows**) to data_cleaned.csv.

2 02_Feature Engineering

- Expands dataset from **8 to 39 columns**.
- **Temporal:** Year, Month, Hour, Day_Period, Season, IsPeakHour (5-10 PM).
- **Metrics:** Sub_metering_total, Apparent_Power, Power_Factor, Consumption_Level.

3 03_Normal Visualizations

- Aggregates active/reactive power & sub-metering data.
- Temporal scales: **Hourly, Daily, Monthly, Yearly, Seasonal**.
- Visualizes trends and home meter rates via Matplotlib.

4 04_Dashboard (Energy Command Center)

- Interactive command center built using Plotly.
- **Key KPIs:** Avg Active Power (1.09 kW), Peak (11.12 kW), Avg Voltage (240.84 V), Avg Current (4.63 A).
- Multi-panel trends for time, patterns & appliance groups.

Thank You!

Questions & Discussion | Household Energy Consumption Analysis